

Companion to Bird & de Moor, Algebra of Programming

§5. Datatypes in Allegories

§5.1. Relators

(5.1a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

(5.1b)

the statement

$\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$. Lemma 5.1

Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$. Theorem 5.1 the book's reverse direction (preserving $\circ \implies$ relator) needs the source to be **Rel**: the counterexample over a tabular allegory that is not **Rel** shows it can fail

$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.

Two relators agreeing on maps are equal. Corollary 5.1

$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$ Ex 5.2

$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly. Ex 5.2

$\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$ Ex 5.5

The **power relator** $P \multimap x \text{ s } P(R) \text{ y s} \iff (\forall a \in x \text{ s. } \exists b \in y \text{ s. } a R b) \wedge (\forall b \in y \text{ s. } \exists a \in x \text{ s. } a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§5.2. Relational products

(5.1)

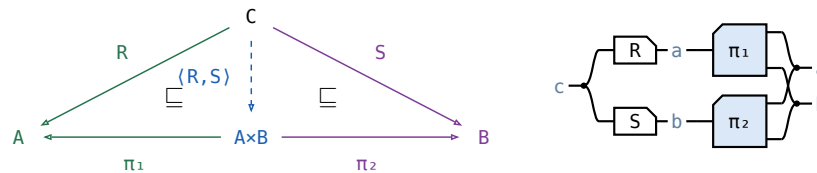
The **fork** of $R : \mathcal{C} \rightarrow \mathcal{A}$ and $S : \mathcal{C} \rightarrow \mathcal{B}$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

(5.2a)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad (5.6)$$

$$\langle R, S \rangle \pi_2 = \text{dom}(R) S \quad (5.7)$$

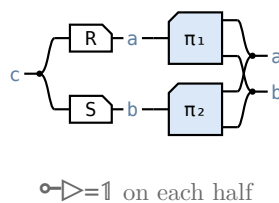
(5.2b)



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In **Rel**, $c \langle R, S \rangle (a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

No $\circ$ survives the translation. $\pi_1 = 1 \otimes \bullet$ discards the second component, so $\pi_1^\circ = 1 \otimes \bullet$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(5.2c)

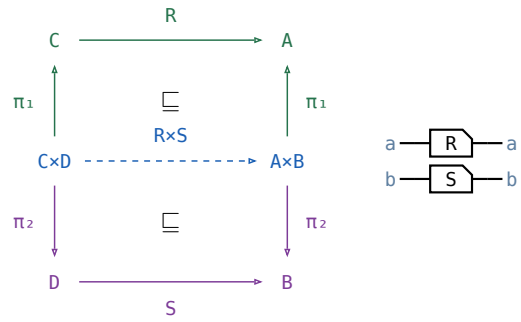


§5.2.1. Relational product $R \times S$

(5.2)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.

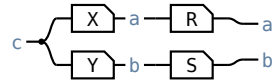
(5.2.1a)



Right-then-up is $(R \times S)\pi_1$, up-then-right is $\pi_1 R$, and $(R \times S)\pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§5.2.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$. Both sides are this picture:



(5.3)

(5.2.2a)

the law	picture
1 $\langle X, Y \rangle (R, S)^\circ = XR^\circ \cap YS^\circ$	
2 $\langle R, S \rangle^\circ (U, V) \sqsubseteq (R^\circ U) \times (S^\circ V)$	
3 $\text{Map}(f) \Rightarrow f \langle R, S \rangle = \langle fR, fS \rangle$ f a map; it fails for an arbitrary arrow;	
4 $F(R \times S) \text{unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$; only $\sqsubseteq$ for an arbitrary R, S	
5 $g = \text{curry}(f) \iff (g \times 1) \text{eval} = f$ reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?	

§5.3. Relational coproducts

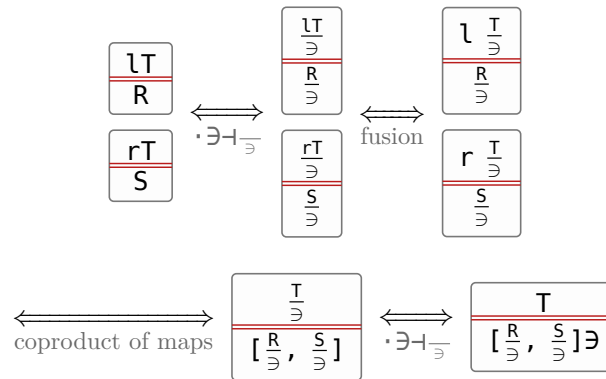
(5.3a)

the statement	picture
$[R, S] \triangleq l \circ R \cup r \circ S$ (5.9) The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$	
$R + S \triangleq [Rl, Sr]$ (5.10)	
$l[R, S] = R, r[R, S] = S$ $lT = R \wedge rT = S \implies T = [R, S]$	
$ll^\circ = 1, rr^\circ = 1$ Ex 5.12	
$lr^\circ = 0, rl^\circ = 0$ Ex 5.12	
$l^\circ l \cup r^\circ r = 1$ Ex 5.12	
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$ (5.11)	

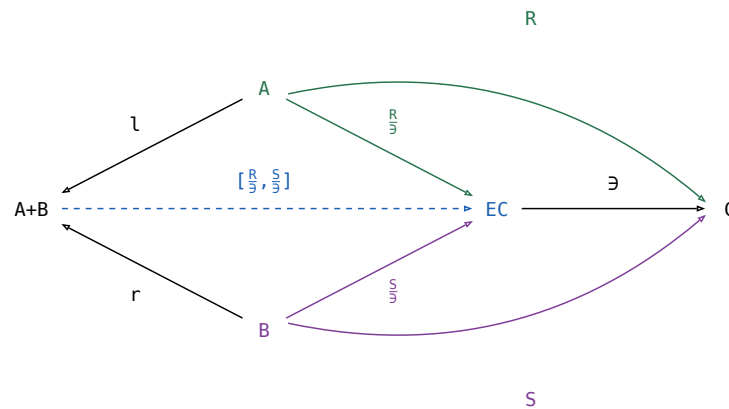
§5.3.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

The universal-property row is not free: l, r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\exists$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow EC$ reversibly, so the map coproduct can be applied underneath it. For any $T : A+B \rightarrow C$,

(5.3.1a)



(5.3.1b)



(5.3a)

Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\frac{R}{\exists} \cup \frac{S}{\exists}) \exists = \frac{R}{\exists} \cup \frac{S}{\exists}$.

(5.3.1c)

the law	picture
prove (5.11), and say why duality does not carry it over from the product law	
$(R+S) \cap [P, Q] [U, V]^\circ =$ $(R \cap P U^\circ) + (S \cap Q V^\circ)$ $[P, Q] [U, V]^\circ$ is a full 2×2 of composites; $R+S$ is diagonal, so the meet cuts the two off-diagonal branches	

§5.4. The power relator $P(R)$

(5.4a)

For $R : A \rightarrow B$,

$$P(R) \triangleq (\epsilon \setminus R \epsilon) \cap (\exists R / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^\circ$$

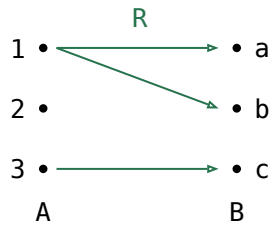
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of xs is related by R to some element of ys , and conversely.

(5.4b)

	in Rel	in words	$\text{im}(xs)$	partner
$(\exists R) / \exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$	$ys \subseteq \text{im}(xs)$	every y has a partner
$(\exists / (\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	every y with some $xs \ R \ y$ is in ys	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{every } y \text{ with some } xs \ R \ y$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
$((\exists R^\circ) / \exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall x. x \ R \ \text{some } ys$	—	every x has a partner
$P(R)$	$\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{some } ys$	—	every x and every y has a partner

(5.4c)



xs	$(\exists R) / \exists$	$(\exists / (\exists R))^\circ$	$E(R)$	$((\exists R^\circ) / \exists)^\circ$	$P(R)$
$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

(5.4d)

law	the reading
$X \sqsubseteq P(R) \iff X \sqsubseteq \exists R \text{ and } X^\circ \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
$P(1) = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$\text{Map}(f) \implies P(f) = E(f)$	In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f(a)$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(RS) = P(R)P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§5.4.1. Type relator

(5.4.1a)

Let F be a binary relator with initial type (α, T) , so T is a type functor. $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the reduce section. For every object A the initial algebra is $\alpha : F(A, TA) \rightarrow TA$, among the maps. T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1) \alpha)$$

(5.4.1b)

name	law	what it says
the defining equation	$T(R) = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way.
type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

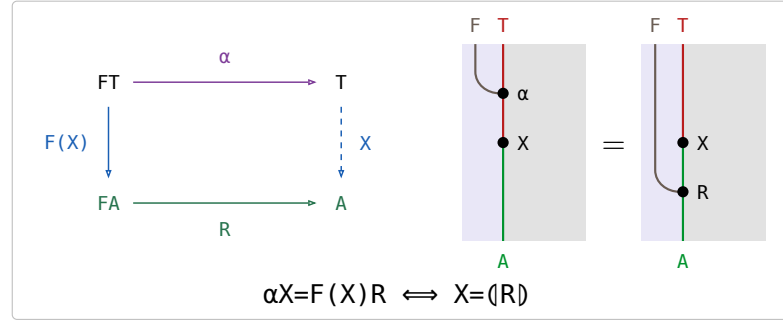
§5.5. Relational catamorphisms

(5.5a)

let F be a relator and has **initial algebra** $\alpha : F(T) \rightarrow T$ in the subcategory of functions. α is also initial in the allegory:

§5.5.1. The defining equation

(5.5.1a)



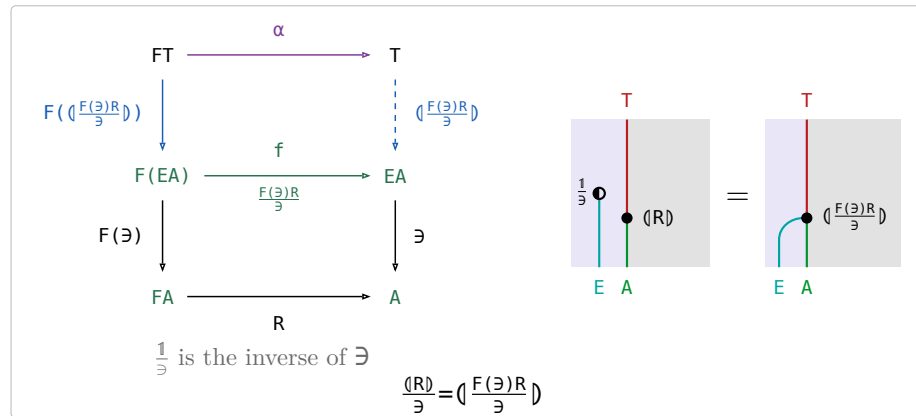
An F -algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $FX \rightarrow X$ at every object and a commuting square at every arrow, but F -Algebra only need it works on T and A .

(5.5.1b)

$\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$		
catamorphism: arrows of the form $\llbracket f \rrbracket$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing		
part	Nat	[A]
datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$
base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
the fold, pointwise	$\llbracket [c, f] \rrbracket(\text{zero}) = c$ $\llbracket [c, f] \rrbracket(\text{succ}(n)) = f(\llbracket [c, f] \rrbracket(n))$	$\llbracket [c, f] \rrbracket(\text{nil}) = c$ $\llbracket [c, f] \rrbracket(\text{cons}(a, x)) = f(a, \llbracket [c, f] \rrbracket(x))$

§5.5.2. $(R) = (\llbracket \frac{F(\exists)R}{\exists} \rrbracket) \exists$

(5.5.2a)



(5.12)

$$\begin{array}{c}
 \boxed{\frac{\alpha X}{F(X)R}} \iff \boxed{\frac{\alpha X}{\frac{F(X)R}{\exists}}} \iff \boxed{\frac{\alpha X}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
 \iff \boxed{\frac{\alpha \frac{X}{\exists}}{F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}}} \iff \boxed{\frac{\frac{X}{\exists}}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket}} \iff \boxed{\frac{X}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists}}
 \end{array}$$

relator, fusion twice reduce of maps $\cdot \exists \dashv \frac{1}{\exists}$

the last two rows at $X := (R)$:

$\alpha X = F(X)R \iff \frac{X}{\exists} = (\frac{F(\exists)R}{\exists})$ (5.5.2b)		
in a tabular allegory, with F a relator and $\alpha : F(T) \rightarrow T$ its initial algebra: a relation X out of T satisfies the fold equation $\alpha X = F(X)R$ of R exactly when $\frac{X}{\exists}$ is the fold of the map $\frac{F(\exists)R}{\exists}$		
$\frac{X}{\exists}$	$\frac{\alpha^\circ F(X)R}{\exists}$	$\frac{\alpha^\circ F(X)R}{\exists}$
 (a)	 (b)	 (c)
 (d)	 (e)	 (f)
(a)	$\frac{X}{\exists}$	α iso; $\alpha X = F(X)R$
(b)	$\frac{\alpha^\circ F(X)R}{\exists}$	α° is a map because α is an iso, so it leaves Λ : $\frac{\alpha^\circ F(X)R}{\exists} = \alpha^\circ$ $\frac{F(X)R}{\exists}$ — pow-laws
(c)	$\alpha^\circ \frac{F(X)R}{\exists}$	$X = \frac{X}{\exists}$ — pow-laws
(d)	$\alpha^\circ \frac{F(\frac{X}{\exists})R}{\exists}$	$F(\frac{X}{\exists}) = F(\frac{X}{\exists})F(\exists)$ as F is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves Λ — relator, pow-laws
(e)	$\alpha^\circ F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}$	fold uniqueness, $\frac{F(\exists)R}{\exists}$ being a map — initial-defn
(f)	$(\frac{F(\exists)R}{\exists})$	

§5.6. Combinatorial functions

definition	type	note
$[A] ::= \text{nil} \mid \text{cons}(A, [A])$	$A \rightarrow A$	The list type, under the short name it keeps.
$\text{list}(R) = (\text{nil}, (R \times 1) \text{cons})$	$[A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.
$\text{subseq} = (\text{nil}, \text{cons} \cup \pi_2)$	$[A] \rightarrow [A]$	$xs \text{ subseq } ys$: ys is xs with elements dropped — cons keeps the head, π_2 drops it.
$\text{prefix} = (\text{nil}, \text{nil} \cup \text{cons})$ $\text{prefix} = \text{cat} \circ \pi_1 = \text{init}^*$	$[A] \rightarrow [A]$	ys is an initial segment of xs ; the first nil is where it stops early. $\text{init} \triangleq \text{snoc} \circ \pi_1$.
$\text{suffix} = \text{cat} \circ \pi_2 = \text{tail}^*$	$[A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons} \circ \pi_2$; as a reduce it needs snoc -lists.
$\text{partition} = \text{concat}^\circ$	$[A] \rightarrow [[A]]$	This cat is restricted to $[A]^* \times [A] \rightarrow [A]$, so ys is a list of non-empty segments of xs .
$\text{concat} = (\text{nil}, \text{cat})$	$[[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.
inits	$[A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\exists}$, listing the prefixes by increasing length.
tails	$[A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.

§5.6.1. $\frac{\text{subseq}}{\exists} = (\text{nil} \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2) \text{cup})$

(5.6.1a)

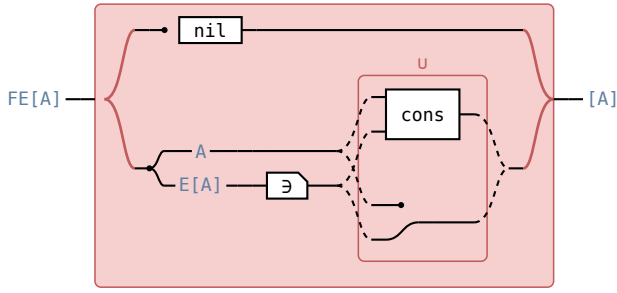
$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : EA \times EA \rightarrow EA$, so $\frac{R \cup S}{\exists} = (\frac{R}{\exists}, \frac{S}{\exists}) \text{cup}$.

$$\frac{F(\exists)[\text{nil}, \text{cons } u \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}] \quad (5.6.1b)$$

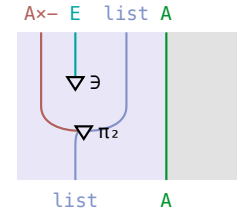
the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12) at **subseq** = $([\text{nil}, \text{cons } u \pi_2])^\exists$, $\triangleq$ **subseq**.

circuit — the fork is $F([A]) = 1 + A \times [A]$: **nil** above, the pair below

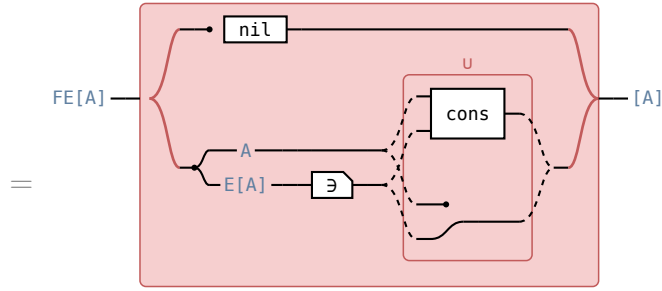
Hinze–Marsden



$$\frac{F(\exists)[\text{nil}, \text{cons } u \pi_2]}{\exists}$$

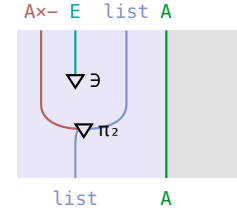


the π_2 operand of $\text{CONS } u \pi_2$ under the $1 \times \exists$ summand of $F(\exists)$, i.e. $(1 \times \exists) \pi_2$

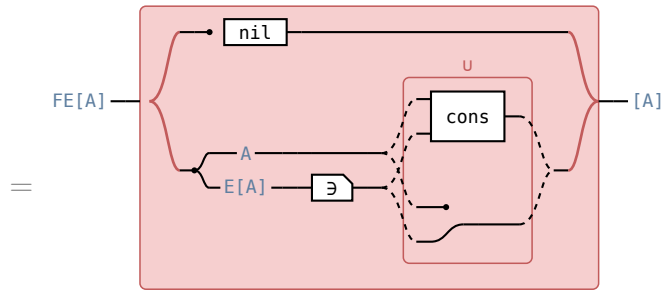


$$\frac{(1+1 \times \exists)[\text{nil}, \text{cons } u \pi_2]}{\exists}$$

$$F(X) = 1 + A \times X \triangleq \text{subseq}$$

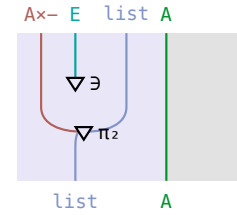


the same operand under $1 + 1 \times \exists$, whose $1 \times \exists$ summand it sits in

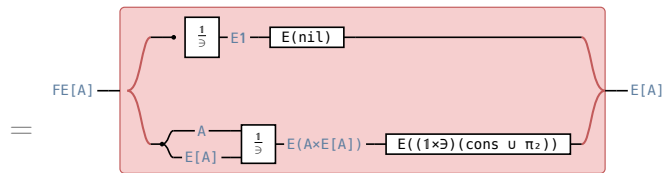


$$\frac{[\text{nil}, (1 \times \exists)(\text{cons } u \pi_2)]}{\exists}$$

$$R + S \triangleq [Rl, Sr], l[R, S] = R, r[R, S] = S \quad (5.3a)$$

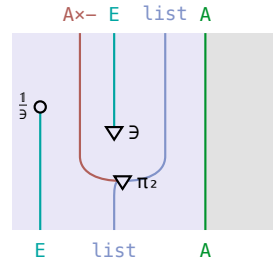


the π_2 operand of the second arm $(1 \times \exists)(\text{cons } u \pi_2)$

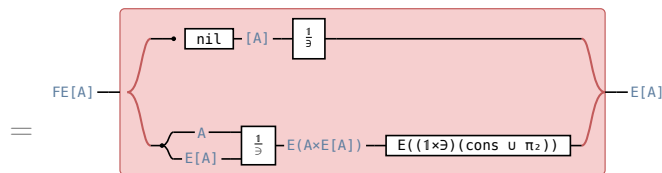


$$\frac{[\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}]}{\exists}$$

$$(5.3.1a) \text{ at } T := [\text{nil}, (1 \times \exists)(\text{cons } u \pi_2)]$$

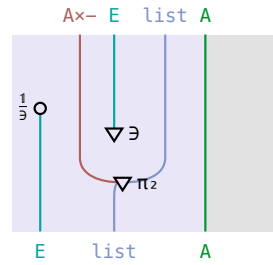


the π_2 operand under its $1 \times \exists$, the arm (5.6.1c)'s square rewrites, $(1 \times \exists) \pi_2 = \pi_2 \exists$



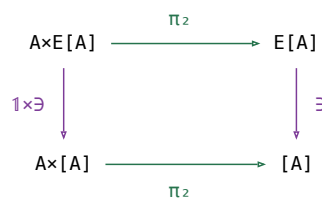
$$\frac{[\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}]}{\exists}$$

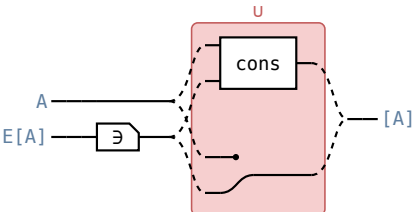
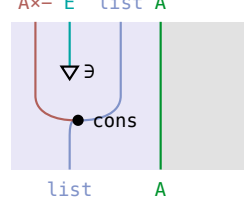
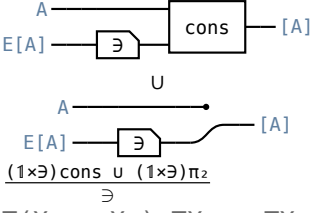
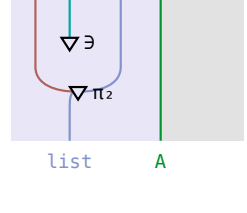
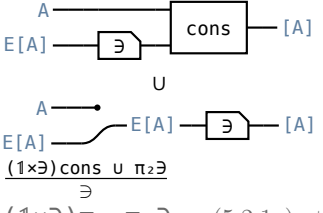
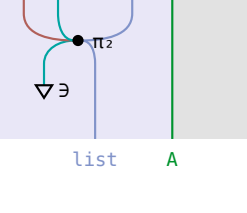
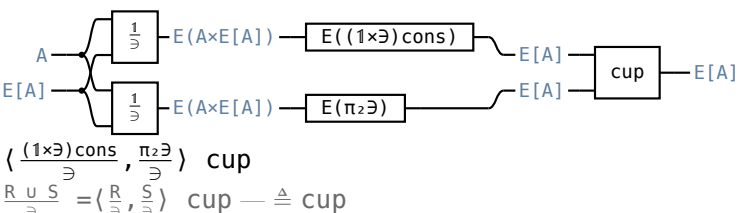
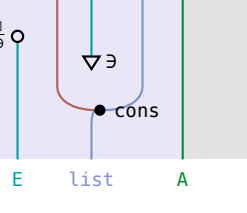
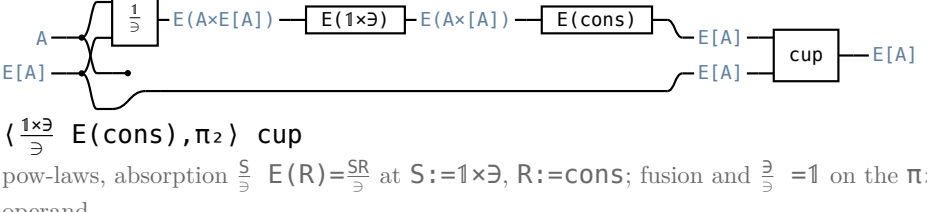
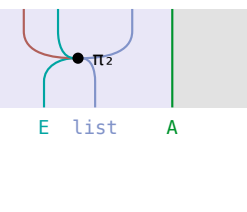
$$\text{pow-laws, } \frac{f}{\exists} = f \frac{1}{\exists} \text{ for } f \text{ a map, at } f := \text{nil}$$



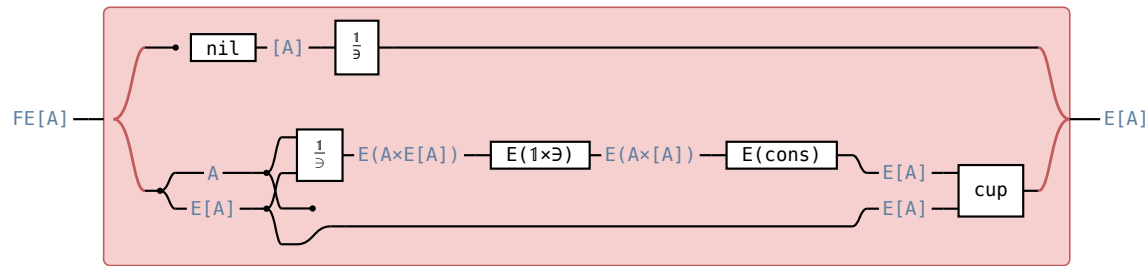
the same operand; the two rows differ only in the **nil** arm

(5.6.1c)



$\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists} = \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ power transpose of join: the power transpose of the join of two relations is $\left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}$, where cup is the function that returns the union of two sets	(5.6.1d)
circuit	Hinze–Marsden
 $\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists}$ (5.6.1b)'s second branch	 the cons operand of $\text{cons} \cup \pi_2$
 $\frac{(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2}{\exists}$ $T(X_1 \cup X_2) = TX_1 \cup TX_2$ — adj-cross	 the π_2 operand of $(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2$
 $\frac{(1 \times \exists)\text{cons} \cup \pi_2 \exists}{\exists}$ $(1 \times \exists)\pi_2 = \pi_2 \exists$ — (5.2.1a) at π_2, an equality because 1 is entire	 the π_2 operand of $(1 \times \exists)\text{cons} \cup \pi_2 \exists$
 $\left(\frac{1 \times \exists}{\exists} \text{cons}, \frac{\pi_2 \exists}{\exists} \right) \text{cup}$ $\frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup} \triangleq \text{cup}$	 the cons operand under its $1 \times \exists$
 $\left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ pow-laws, absorption $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$ at $S := 1 \times \exists$, $R := \text{cons}$; fusion and $\frac{\exists}{\exists} = 1$ on the π_2 operand	 the π_2 operand, bare π_2

(5.6.1e)



$$\left[\frac{\text{nil}}{\exists}, \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \right]$$

which writes $P\text{cons}$; cons is a map, and there $P(\text{cons}) = E(\text{cons})$ — (5.4d).

§5.7. Lax natural transformations (LaT)

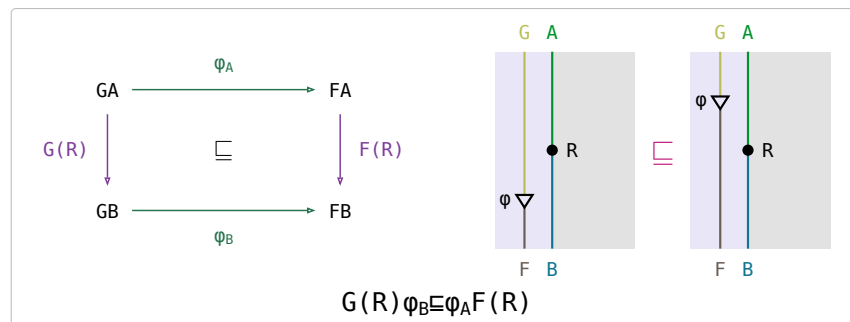
(5.7a)

For relators $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and components $\varphi_A : GA \rightarrow FA$, φ is **lax at** $R : A \rightarrow B$ when $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ — both sides $GA \rightarrow FB$, the left through the component at B , the right through the one at A .

$\varphi : G \Rightarrow F$ is a **lax natural transformation** (LaT) when it is lax at **every** R . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality $G(f)\varphi = \varphi F(f)$: laxness is about relations only.

(5.7b)



closed under	commutative diagram	string diagram
composition $\psi\phi$	$H(R)\psi_B \sqsubseteq \psi_A G(R)$ and $G(R)\phi_B \sqsubseteq \phi_A F(R)$ give $H(R)(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A)F(R)$	
horizontal composition $\chi\circ\phi$	$\phi : G \Rightarrow F$ with $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and $\chi : L \Rightarrow K$ with $L, K : \mathcal{D} \rightarrow \mathcal{E}$ give $\chi \circ \phi : L \circ G \Rightarrow K \circ F$, the family $A \mapsto \chi_{GA} K(\phi_A)$ $L(G(R))\chi_{GB} \sqsubseteq \chi_{GA} K(G(R))$ is χ lax at $G(R)$; then $K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))$ is K applied to ϕ's own inequation; $\chi := \mathbb{1}_K$ gives $K(\phi) : K \circ G \Rightarrow K \circ F$, K composed on the outside; $\phi := \mathbb{1}_G$ gives $\chi G : L \circ G \Rightarrow K \circ G$, G composed on the inside two candidates, $\chi_{GA} K(\phi_A)$ and $L(\phi_A)\chi_{FA}$, ordered by $\sqsubseteq$; this row is the first	
union $\phi \cup \psi$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ and $G(R)\psi_B \sqsubseteq \psi_A F(R)$ give $G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A)F(R)$	
a relator K $K(\phi)$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ gives $K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))$	
product $\phi \times \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi \times \psi : G \times G' \Rightarrow F \times F'$ $(R \times S)(U \times V) = (RU) \times (SV)$ and monotonicity in both slots — the row above's K applied to the inequation, run on a bifunctor; the fork is the derived case $\langle \phi, \psi \rangle = \triangleleft(\phi \times \psi)$, and its ONE cost is the lax copy law $R \triangleleft \triangleleft (R \times R)$ — rel-monoid	
coproduct $\phi + \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi + \psi : G + G' \Rightarrow F + F'$ $(R + S)(U + V) = (RU) + (SV)$ and monotonicity in both slots; the co-fork is the derived case $[\phi, \psi] = (\phi + \psi) \triangleright$, and $\triangleright$ costs nothing the step it would need is $(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)$, the wrong direction of meet-semidistrib	
meet — fails $\phi \cap \psi$	a counterexample: (5.7.1a)	

Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A(XY) \sqsubseteq (X'Y')R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \phi_B, \phi_A$, quantified over R ; at $X' = X$ with R_A, R_B endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference

